WORKED SOLUTION

STEP II 2001, Question 7

Pure • Complex Numbers and Rotations

STEP II 2001, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q7

The key idea

Walking round a regular polygon, every side has the same length and at every vertex we turn through the same *exterior* angle. In the Argand diagram a turn through angle θ is multiplication by $e^{i\theta}$, so each side is the previous one multiplied by a fixed complex number. Once the vertices are known, the centre comes from a perpendicular bisector, and the rotation is just one more multiplication by $e^{i\theta}$.

Solution

Finding Q and S

The interior angle at each vertex is $5\pi/6$, so at each vertex the direction of travel turns through the exterior angle $\pi - 5\pi/6 = \pi/6$. Since the polygon lies in the upper half-plane and we start along the positive real axis from O to P, the turns must be *anticlockwise* (a clockwise turn at P would take Q below the real axis).

Write $\omega = e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The side $\overrightarrow{OP}$ is the complex number 2, and each later side is the previous one multiplied by ω (same length, turned by $\pi/6$). So

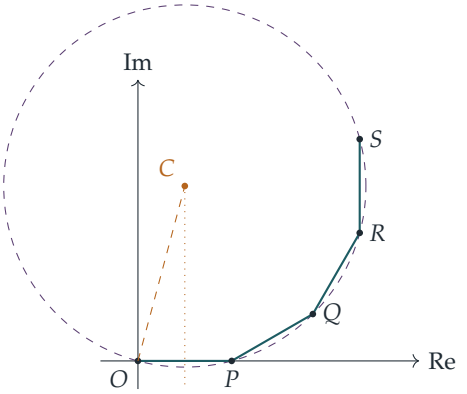
$$\begin{aligned}\overrightarrow{PQ} &= 2\omega = \sqrt{3} + i,\\ \overrightarrow{QR} &= 2\omega^2 = 2e^{i\pi/3} = 1 + \sqrt{3}\,i,\\ \overrightarrow{RS} &= 2\omega^3 = 2e^{i\pi/2} = 2i.\end{aligned}$$

Adding up,

$$\begin{aligned}Q &= 2 + (\sqrt{3} + i) = 2 + \sqrt{3} + i,\\ R &= Q + 1 + \sqrt{3}\,i = (3 + \sqrt{3}) + (1 + \sqrt{3})i,\\ S &= R + 2i = (3 + \sqrt{3}) + (3 + \sqrt{3})i.\end{aligned}$$

The first line is the required value of Q, and

$$S = (3 + \sqrt{3})(1 + i).$$



The centre C

The circle through O, P, Q is the circumcircle of the whole 12-gon, so C is the centre of the polygon. Since C is equidistant from O and $P = 2$, it lies on the perpendicular bisector $\operatorname{Re} z = 1$; write $C = 1 + h\mathrm{i}$. It is also equidistant from O and Q:

$$\begin{aligned} |C|^2 = |C - Q|^2 &\iff 1 + h^2 = (1 + \sqrt{3})^2 + (h - 1)^2 \\ &\iff 1 + h^2 = 4 + 2\sqrt{3} + h^2 - 2h + 1 \\ &\iff 2h = 4 + 2\sqrt{3}, \end{aligned}$$

so $h = 2 + \sqrt{3}$ and $C = 1 + (2 + \sqrt{3})\mathrm{i}$, in the first quadrant.

Ask yourself What is the argument of C exactly?

We need $\tan(\arg C) = 2 + \sqrt{3}$. Now

$$\tan \tfrac{5\pi}{12} = \tan\left(\tfrac{\pi}{4} + \tfrac{\pi}{6}\right) = \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} = \frac{(\sqrt{3} + 1)^2}{2} = 2 + \sqrt{3},$$

and C is in the first quadrant, so $\arg C = 5\pi/12$. (Geometrically: the triangle OCP is isosceles with apex angle $2\pi/12$ at C, so its base angles are $\frac{1}{2}(\pi - \pi/6) = 5\pi/12$.)

The rotation

Rotating anticlockwise about O by angle θ multiplies every point by $\mathrm{e}^{\mathrm{i}\theta}$, so it moves C to argument $5\pi/12 + \theta$. For C to lie on the real axis we need $5\pi/12 + \theta$ to be a multiple of π; the smallest positive θ is

$$\theta = \pi - \tfrac{5\pi}{12} = \tfrac{7\pi}{12},$$

which puts C on the *negative* real axis.

Now write S in modulus–argument form: $1 + \mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$, so $S = (3 + \sqrt{3})\sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4} =$

$(3\sqrt{2}+\sqrt{6})\,\mathrm{e}^{\mathrm{i}\pi/4}$. The new position of S is

$$\begin{aligned} S' = S\,\mathrm{e}^{7\mathrm{i}\pi/12} &= (3\sqrt{2}+\sqrt{6})\,\mathrm{e}^{\mathrm{i}(3\pi/12+7\pi/12)} = (3\sqrt{2}+\sqrt{6})\,\mathrm{e}^{5\mathrm{i}\pi/6} \\ &= (3\sqrt{2}+\sqrt{6})\left(-\tfrac{\sqrt{3}}{2}+\tfrac{1}{2}\mathrm{i}\right) = -\tfrac{1}{2}(3\sqrt{2}+\sqrt{6})(\sqrt{3}-\mathrm{i}), \end{aligned}$$

as required.

Checks

- $|C|^2 = 1+(2+\sqrt{3})^2 = 8+4\sqrt{3} = (\sqrt{6}+\sqrt{2})^2$, and $|C-S|^2 = (2+\sqrt{3})^2+1$ too, so S really is on the same circle.
- $|S'| = 3\sqrt{2}+\sqrt{6} = |S|$, as a rotation must preserve.
- Numerically $S \approx 4.732+4.732\mathrm{i}$ and $S' \approx -5.7956+3.3461\mathrm{i}$; the computer confirmed all the side lengths, the angles of $5\pi/6$, and that $7\pi/12$ is the first anticlockwise angle putting C on the real axis.

Common slips

- Turning by the interior angle $5\pi/6$ instead of the exterior angle $\pi/6$.
- Rotating by $-5\pi/12$ (clockwise) or by $\pi/2-5\pi/12$: the rotation is anticlockwise, and C first meets the real axis on its negative half.

Beyond STEP

Multiplication by $\mathrm{e}^{\mathrm{i}\theta}$ as rotation is the start of seeing complex functions as geometric transformations: $z \mapsto az+b$ gives all rotations, enlargements and translations of the plane, and these form a group. The vertices of a regular polygon are a geometric sum, $2(1+\omega+\cdots+\omega^{k-1})$, linking to the roots of unity. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.